PROJECT SUBMISSION REPORT

Inventory & Sales Management System with Analytics Dashboard
for GreenLeaf Grocers

Submitted by	Nishu Sehrawat
Education	B.Sc. — Third Year
College	Government Dungar College, Bikaner
Project area	Data Analytics
Tools	Python, pandas, CSV data, charts/dashboard tools

1. Introduction About Me

My name is Nishu Sehrawat, and I am a third-year B.Sc. student at Government Dungar College, Bikaner. I am developing my skills in data analytics and learning how to use data to understand problems and support better decisions. I am building my foundation in tools such as Python, SQL, Excel, Power BI, pandas, and data visualization. Through practical projects, I aim to improve my analytical thinking, problem-solving, and ability to explain insights clearly.

2. Project Introduction

The project, “Inventory & Sales Management System with Analytics Dashboard for GreenLeaf Grocers,” focuses on analyzing retail sales and monitoring inventory. Sales and stock data can help a business understand which products and categories perform well, how revenue changes over time, and which items may need to be restocked.

3. Problem Statement

A grocery business may find it difficult to track sales performance and stock levels by looking at individual records. Without a clear summary, it can be harder to compare categories, identify sales trends, and notice products that are approaching their reorder point. This project proposes a simple data-analysis workflow and dashboard to make those patterns easier to review.

4. Project Objectives

1. Calculate total revenue and total units sold.

2. Compare revenue across product categories and regions.

3. Review monthly sales trends.

4. Identify records where closing stock is at or below the reorder level.

5. Present important measures in a clear, easy-to-understand dashboard.

5. Dataset Description

The project CSV contains fields such as order ID, order date, product name, category, units sold, unit price, revenue, opening stock, reorder level, closing stock, supplier, and region. These columns support sales summaries

and basic inventory checks.

Dataset note: The image provided for this task showed the project brief but did not include an original data file. The companion CSV created for this package is a synthetic practice dataset, not actual company data. If an official dataset was provided separately by the instructor, it should replace the sample CSV and the analysis should be recalculated.

6. Methodology / Steps

1. Data loading: Read the CSV file using pandas.

2. Data checking: Inspect the rows and columns, check missing values, and look for duplicate order IDs.

3. Preparation: Convert order dates to date format and create month fields for time-based analysis.

4. Sales analysis: Calculate revenue and units sold, then group results by category, region, and month.

5. Inventory check: Compare closing stock with the reorder level to flag records that may need replenishment.

6. Visualization: Use bar charts, line charts, KPI cards, or a dashboard to communicate the results.

7. Expected Outcome

The expected outcome is a clear overview of sales and inventory information. The analysis should help a user compare product categories, review changes in revenue over time, and identify low-stock records that may require attention. The dashboard is intended to make the information easier to understand than a raw spreadsheet alone.

8. Short Task Explanation

In this task, I am using data analytics to study sales and inventory for a grocery business. First, I load and check the data. Then I calculate revenue and units sold, compare sales by category and region, and examine monthly trends. I also check stock levels against reorder levels to identify items that may need replenishment. Finally, I present the important findings using charts or a dashboard. This shows how organized data can support everyday business decisions.

9. Conclusion

This project demonstrates a practical use of data analytics in retail. By summarizing sales and monitoring stock levels, a business can get a clearer view of its performance and identify areas that need attention. The project also provides an opportunity to practise data cleaning, analysis, and visualization.

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